

Roll No.

Total No. of Pages : 03

Total No. of Questions : 16

BCA (2014 to 2018) / B.Sc. (IT) (2015 to 2018) (Sem.-1)

MATHEMATICS-I

Subject Code : BSIT/BSBC-103

M.Code : 10045

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains SIX questions carrying TEN marks each and students have to attempt any FOUR questions.

SECTION-A

Write briefly :

- Q1. If A, B, C are any sets, prove that $A - (B \cup C) = (A - B) \cap (A - C)$.
- Q2. Define partition of sets.
- Q3. Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. The family $\{\{1, 4, 8\}, \{3, 5, 9\}, \{2, 7\}, \{6, 10\}\}$ is a partition of X. Determine the equivalence relation corresponding to the above partition.
- Q4. Let $X = \{1, 2, 3, 4\}$ and $R = \{(x, y) : x > y\}$. Draw the diagram and matrix of R.
- Q5. Using truth table, prove that $\sim(p \rightarrow q) = p \wedge \sim q$.
- Q6. Given the proposition over the natural numbers $p: n < 4$, $q: 2n > 17$ and $r: n$ is a divisor of 18. What are the truth sets of $p \wedge q$ and $q \rightarrow r$.
- Q7. Prove that the number of edges in a complete graph with n vertices is $\frac{n(n-1)}{2}$.
- Q8. Draw a simple planar graph with 6 nodes and 11 edges.
- Q9. Define recurrence relation with example.
- Q10. Solve the recurrence relation $S(K) - S(K - 1) - S(K - 2) = 0$.

SECTION-B

- Q11. a) State and prove De Morgan's Laws for sets.
- b) The relation R is defined by $(a, b) \in R$ if and only if 5 divides $b - a$. **Show that R is an equivalence relation.**
- Q12. a) Let $R = \{(a, b): |a - b| = 1\}$ and $S = \{(a, b): a - b \text{ is even}\}$ are two relations on $A = \{1, 2, 3, 4\}$. Then
- Find matrices of R and S .
 - Draw diagrams of R and S
 - Using matrices of R and S , find the relation RS .
- b) Test the validity of "*If my brother stands first in the class, I will give him a watch. Either he stood first or I was out of station. I did not give my brother a watch this time. Therefore I was out of station.*"
- Q13. a) Over the universe of Books, define the proposition $B(x)$: x has a blue cover, $M(x)$: x is a mathematics book, $U(x)$: x is published in United Estate and $R(x, y)$: The bibliography of x includes y .

Translate into words :

- $(\exists x)(M(x) \wedge \sim B(x))$.
- $(\forall x)(M(x) \wedge U(x) \rightarrow B(x))$
- $(\exists x)(\sim B(x))$

Express using quantifiers :

- Every book with blue cover is a mathematics book.
 - There are mathematics books that are published outside the United States.
 - Not all books have bibliography.
- b) Use Mathematical Induction to show that $1+2+4+\dots+2^n = 2^{n+1}-1$.

Q14. Using Dijkstra's Algorithm, find shortest path from A to D.

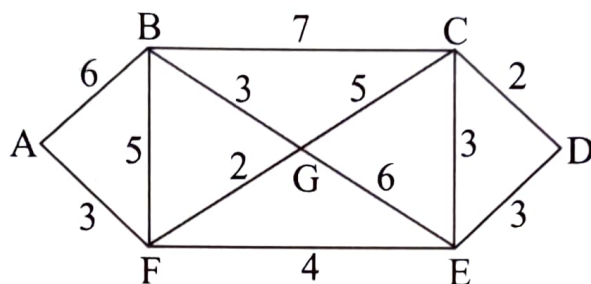


FIG. 1

Q15.a) Find the minimal spanning tree for the following weighted connected graph using Kruskal's Algorithm.

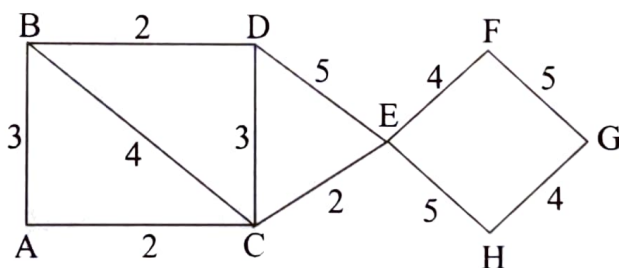


FIG. 2

b) Solve $S(K) - 2S(K-1) + S(K-2) = 0$, where $S(0) = 1$, $S(1) = 2$.

Q16. a) Solve $S(K) - 7S(K-1) + 10S(K-2) = 6 + 8K$, where $S(0) = 1$, $S(1) = 2$.

b) Find inverse of the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 3 \\ 3 & -1 & -1 \end{bmatrix}$.

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Total No. of Pages : 02

Total No. of Questions : 07

B.Sc. (IT / Graphics & Web Designing) / BCA (Sem.-1)

MATHEMATICS

Subject Code : UGCA-1901

M.Code : 76961

Date of Examination : 10-01-2023

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains SIX questions carrying TEN marks each and students have to attempt any FOUR questions.

SECTION-A

1. Write briefly :

- a) Define Power Set with an illustration.
- b) What are tautologies, contradiction and contingent statements in algebra of logic?
- c) Define Conjunction, Disjunction, Conditional Operator and Bi-Conditional Operator.
- d) Define Addition and Multiplication of matrices with an example.
- e) Which term in the A.P. 5, 2, -1, is - 22?
- f) If $1/b+c$, $1/c+a$, $1/a+b$ are in A.P. then prove that a^2 , b^2 , c^2 are in A.P.
- g) In a G.P., the third term is 24 and sixth term is 192. Find the tenth term.
- h) If A, B are any sets, then $A \cap B = B$ if $B \subset A$.
- i) State De-Morgan's Law for difference of sets.
- j) Define tautology with an example.

SECTION-B

2. a) For given $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -1 & 1 \end{bmatrix}$. Is the following statement true or false $(A + B)^2 = A^2 + B^2 + 2AB$.
- b) Find the values of x, y, z and w for given $3 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$.
3. a) Out of 400 boys of a school, 112 played cricket, 120 played hockey and 168 played football of these 32 played both football and hockey, 40 played cricket and football, 20 played cricket and hockey, 12 played all the three games. How many boys did not play any game and how many play only one game?
- b) State and Prove De-Morgan's law of sets.
4. a) Find the value of a_7 from the recurrence relation with $a_n = 2a_{(n-1)} + 3$ with $a_0 = 6$
- b) If a, b, c are in A.P. and b, c, d are in G.P. and $1/c, 1/d, 1/e$ are in A.P. Prove that a, c, e are in G.P.
5. a) If $A = \begin{pmatrix} 2 \\ -4 \\ 5 \end{pmatrix}$ and $B = (6 \ 3 \ -1)$ then verify that $(AB)' = B'A'$
- b) If $A = \begin{pmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{pmatrix}$ Express A as the sum of two matrices such that one is symmetric and other is skew – symmetric.
6. a) If $A = \begin{pmatrix} 1 & 3 & 2 \\ 0 & 5 & 7 \\ 6 & 4 & 8 \end{pmatrix}$ find the value of $A^2 + 7A + 3I$ here I denotes identity matrix.
- b) Prove that it is $A \subset B$ if and only if $B^c \subset A^c$ where c denotes complement of set.
7. a) Prove that $(p \wedge q) \wedge \sim(p \vee q)$ is a fallacy.
- b) Check the validity of the argument :

If I work, I cannot study. Either I work or pass mathematics. I passed mathematics. Therefore, I study.

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SECTION – B

2. a) A college awarded 38 medals in Foot-ball, 15 in basket ball and 20 medals in cricket. If there medals went to a total of 58 men and only three men got medals in all the three sports, how many received medal in exactly two of the three sports?

b) Let $A = \{x : x \text{ is multiple of } 2, x \in \mathbb{N}\}$

$B = \{x : x \text{ is multiple of } 5, x \in \mathbb{N}\}$

$C = \{x : x \text{ is multiple of } 10, x \in \mathbb{N}\}$

Then find $A \cup (B \cap C)$, $(A \cap B) \cap C$, $A \cup (B \cup C)$.

3. a) Test the validity of :

Unless we control population, all advances resulting from planning will be nullified but this must not be allowed to happen. Therefore we must somehow control population.

b) Prove that $[(p \rightarrow q) \times (q \rightarrow r)] \Rightarrow (p \rightarrow r)$ is a tautology,

4. a) If $A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \end{bmatrix}$ and $k_1 = 1, k_2 = 2$ then verify that $(k_1 + k_2) A = k_1 A + k_2 A$.

b) If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & -1 \end{bmatrix}$ then determine A^2 .

5. Prove that an undirected graph possesses a Eulerian circuit if and only if it is connected and has its vertices of even degree.

6. a) Prove that associativity holds over conjunction by using propositional calculus.

b) Solve $S(k) - 7s(k-1) + 10 S(k-2) = 6 + 8k$ with $S(0) = 1$ and $S(1) = 2$.

7. Use the principle of mathematical Induction to prove that

$$1.3 + 2.4 + 3.5 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6} \text{ for any natural number } n.$$

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